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Key Points:

- Nonlinear dispersive simulation with a high-resolution source model
- Reproduction of the records for the focal, offshore, and inundation area
- The roles of the dispersion, nonlinear, and dissipation effects are revealed

Supporting Information:

- Readme
- Movie S1
- Movie S2
- Movie S3
- Movie S4

Correspondence to:
T. Saito,
saito-ta@bosai.go.jp

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Dispersion and nonlinear effects in the 2011 Tohoku-Oki earthquake tsunami

Tatsuhiko Saito[1], Daisuke Inazu[1], Takayuki Miyoshi[2], and Ryota Hino[3]

[1]National Research Institute for Earth Science and Disaster Prevention, Tsukuba, Japan, [2]Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan, [3]Research Center for Prediction of Earthquakes and Volcanic Eruptions, Graduate School of Science, Tohoku University, Sendai, Japan

Abstract This study reveals the roles of the wave dispersion and nonlinear effects for the 2011 Tohoku-Oki earthquake tsunami. We conducted tsunami simulations based on the nonlinear dispersive equations with a high-resolution source model. The simulations successfully reproduced the waveforms recorded in the offshore, deep sea, and focal areas. The calculated inundation area coincided well with the actual inundation for the Sendai Plain, which was the widest inundation area during this event. By conducting sets of simulations with different tsunami equations, we obtained the followings insights into the wave dispersion, nonlinear effects, and energy dissipation for this event. Although the wave dispersion was neglected in most studies, the maximum amplitude was significantly overestimated in the deep sea if the dispersion was not included. The waveform observed at the station with the largest tsunami height ($\sim$2 m) among the deep-ocean stations also verified the necessity of the dispersion. It is well known that the nonlinear effects play an important role for the propagation of a tsunami into bays and harbors. Additionally, nonlinear effects need to be considered to accurately model later waves, even for offshore stations. In particular, including nonlinear terms rather than the inundation was more important when precisely modeling the waves reflected from the coast.

1. Introduction

The Tohoku-Oki earthquake occurred off the shore of northeastern Honshu, Japan, on 11 March 2011. This was the largest event (M_W 9.1) since instrumental observations began in Japan [e.g., *Hirose et al.*, 2011]. A gigantic tsunami excited by the earthquake struck the Pacific coast of Japan and more than 15,000 people were killed [e.g., *Hayashi et al.*, 2011]. The ocean-bottom pressure gauges deployed in the focal area recorded the huge tsunami [*Saito et al.*, 2011] produced by the seafloor rise of $\sim$ 5 m [*Ito et al.*, 2011]. The oncoming huge tsunami was recorded from offshore to onshore along the Sanriku coast [*Fujii et al.*, 2011; *Maeda et al.*, 2011]. The inundation heights and area were investigated widely along the Pacific coast of northeastern Honshu, Japan, by the 2011 Tohoku Earthquake Tsunami Joint Survey Group [e.g., *Mori et al.*, 2011] and also by aerial photographs [*Nakajima and Koarai*, 2011]. The Deep-ocean Assessment and Reporting of Tsunamis (DART) system recorded the tsunami propagation across the Pacific Ocean and it successfully predicted the tsunami arrival for various far-field coasts [*Tang et al.*, 2012]. A number of high-quality data were recorded for this event. With those data sets, the Tohoku-Oki earthquake tsunami has been analyzed from various points of view.

Estimating the tsunami source characteristics is a basic requirement to reveal the nature of a tsunami. The slip distribution on the earthquake fault was examined by tsunami forward modeling [e.g., *Grilli et al.*, 2012, 2013; *Lovholt et al.*, 2012], and estimated by the tsunami waveform inversion analyses [e.g., *Fujii et al.*, 2011; *Satake et al.*, 2013]. Some studies used not only tsunami data but also seismic and geodetic data for the estimation of the earthquake source process [e.g., *Yokota et al.*, 2011; *Gusman et al.*, 2012; *Hooper et al.*, 2012; *Yamazaki et al.*, 2013]. *Saito et al.* [2011] constructed an initial tsunami-height distribution based on the tsunami waveform inversion analysis, in which they used near-field tsunami waveforms that included the records inside the focal area [*Hino et al.*, 2009] and employed the linear Bousinessq equations (linear dispersive equations) to correctly reproduce a short-wavelength tsunami source. Although differences are generally recognized among the source models of those studies, the estimated sources show a common feature in which a steep and high initial-height distribution of more than 8 m

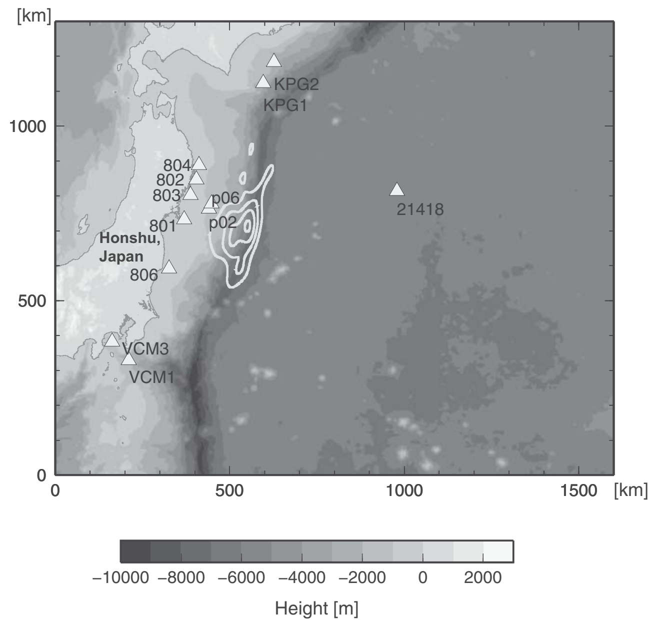

Figure 1. The initial tsunami height distribution for the simulation of the 2011 Tohoku-Oki earthquake and the stations that provided the waveform records used in this study. The initial tsunami height distribution is contoured from 2 to 8 m at intervals of 2 m. Ocean-bottom pressure gauges are equipped at stations KPG1, KPG2, VCM1,VCM3, P02, P06, and 21418, and GPS wave gauges are equipped at stations 801, 802, 803, 804, and 806.

formed near the trench. This steep bump near the trench was caused by a large slip at the very shallow region ($<$ 10 km) of the earthquake fault.

Developing a real-time tsunami forecasting technique is an important subject for mitigating disasters. The Japan Meteorological Agency (JMA) operated a tsunami warning system that was based on the magnitude of the earthquake, which was determined from a seismogram analysis, before the 2011 Tohoku-Oki earthquake occurred [e.g., *Tatehata*, 1997]. However, the analysis failed to accurately estimate the earthquake and tsunami sizes, because the long-period ($> \sim$100 s) seismic waves were not properly included for the rapid magnitude estimation [e.g., *Kanamori*, 2012]. Conversely, according to a tsunami waveform analysis, the tsunami forecasting by the DART system estimated the total tsunami energy within 1.5 h and successfully forecasted the tsunami arrivals and heights in the far-field coasts [*Tang et al*., 2012]. Numerous studies have been conducted to develop tsunami-forecasting techniques for near-field tsunami [e.g., *Tsushima et al*., 2011; *Ohta et al*., 2012; *Gusman and Tanioka*, 2013; *Melgar and Bock*, 2013; *Wei et al*., 2013].

Investigating the propagation properties is the fundamental research topic and work when estimating the source nature and developing a reliable warning system. The dispersive tsunami was observed and investigated for the 2004 Indian Ocean tsunami [e.g., *Kulikov*, 2006]. Simulation studies for dispersive waves were also conducted [e.g., *Horrillo et al*., 2006; *Ioualalen et al*., 2006]. As for the Tohoku tsunami, *Kirby et al*. [2012] reported that oscillatory dispersive tail appeared at the stations located far ($\sim$6200 km) from the source by comparing the observed records and the sets of numerically simulated tsunamis. The Tohoku tsunami was so large that the tsunami reflected from the coast of Chile was clearly recorded at the Japanese coast two days after the occurrence of the earthquake [*Saito et al*., 2013]. The tsunami remained the entire Pacific Ocean and it took more than 6 days for the tsunami to decrease below the background signal level [e.g., *Fine et al*., 2012; *Heidarzadeh and Satake*, 2013; *Rabinovich et al*., 2013]. The tsunami decay process in the Pacific Ocean can be reproduced by an energy dissipation characterized by a bottom friction that is much weaker than those reported for a near-coast tsunami [*Saito et al*., 2013]. In the very shallow sea or near the coasts, nonlinear wave phenomena were observed. At a river mouth in the Sendai plain, soliton fission with a wavelength of a few hundred meters was taken by a video camera from a helicopter. *Murashima et al*. [2012] reproduced the soliton fission using nonlinear dispersive simulations. The characteristics of the inundation strongly depend on the shapes of the bays and the topography around the coast. Numerical simulations were conducted for reproducing and understanding the inundation process of the Tohoku tsunami [e.g., *Shimozono et al*., 2012; *Sugawara et al*., 2012].

In order to clarify the propagation mechanism of the Tohoku tsunami and to deepen our understanding of the wave propagation phenomena, there are two important points to consider. One is the requirement to employ a realistic source model in the tsunami simulations, since the propagation process strongly depends on the source properties. For example, the wave dispersion varies according to how

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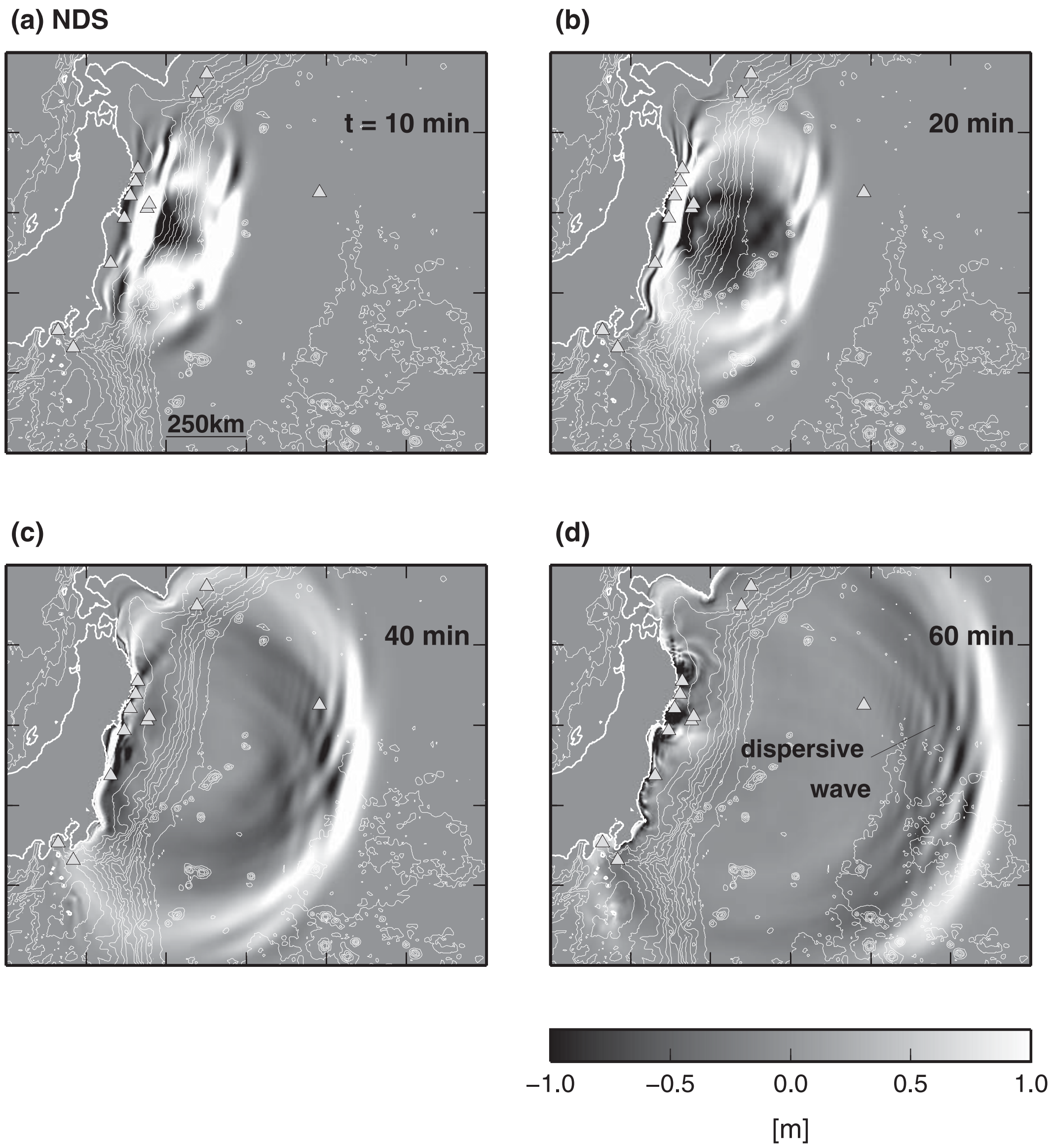

Figure 2. Simulation results of the nonlinear dispersive (NDS) tsunami equations. Tsunami height distribution at the elapsed time of (a) 10, (b) 20, (c) 40, and (d) 60 min from the simulation start time. The sea depth is contoured at intervals of 1000 m.

much short wavelength component the tsunami source contains. Also, the nonlinear features dramatically change the wave field according to its amplitude. Another important point for the study of propagation is to compare the simulation results of various tsunami equations to elucidate which factor (dispersion, nonlinearity, energy dissipation, etc.) is essential to reproduce the observed features when investigating the characteristics for the propagation of a tsunami from the deep ocean to near the coasts.

Which stations observed and which parts of the waveforms of dispersion and nonlinearity were actually observed during the 2011 Tohoku tsunami? These are fundamental questions on the tsunami propagation problem. However, previous studies do not seem to give satisfactory answers to these questions. This is mainly because the purpose of many previous studies was to select the most probable earthquake-faulting scenario, rather than investigate the propagation process for the Tohoku-Oki earthquake. In this study, we intend to reveal the roles of wave dispersion and nonlinear effects as the propagation mechanisms of the 2011 Tohoku-Oki earthquake tsunami. We conducted numerical simulations

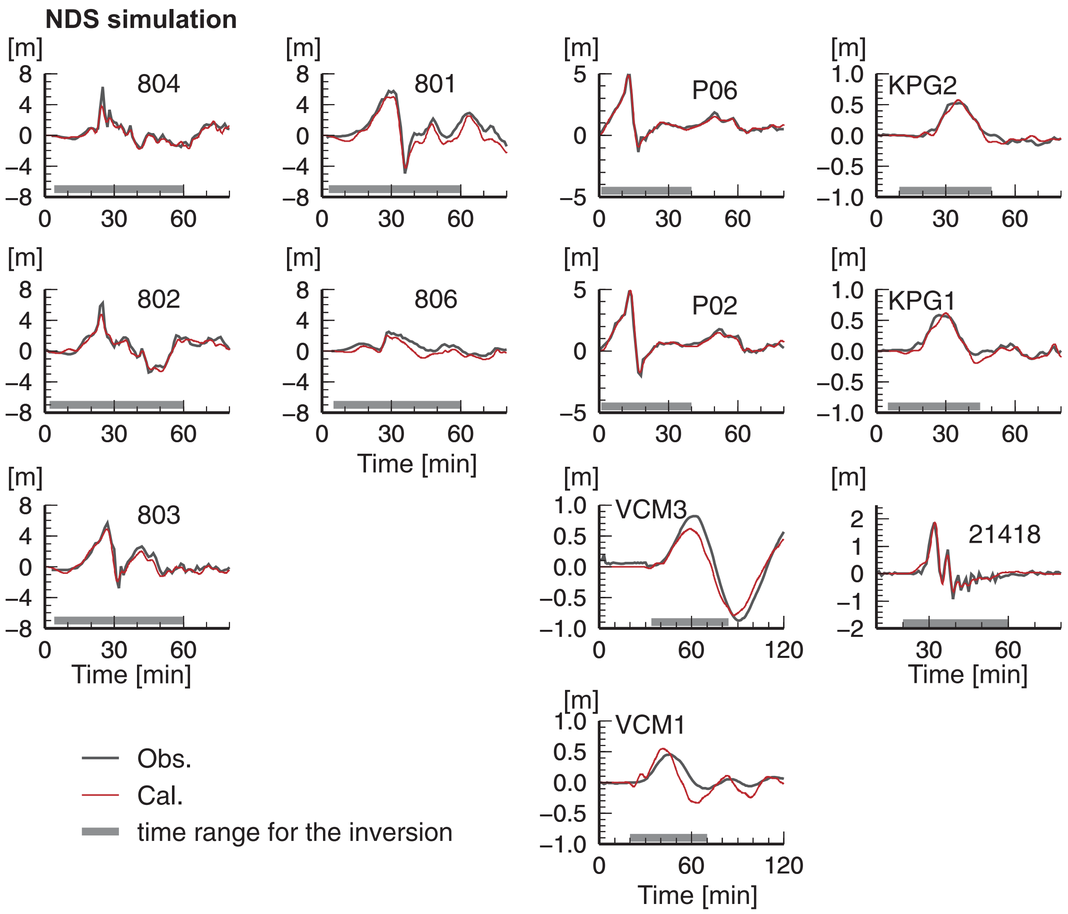

Figure 3. Tsunami waveforms recorded at the stations with the ocean-bottom pressure gauges (KPG1, KPG2, VCM1,VCM3, P02, P06, and 21418) and GPS wave gauges (801, 802, 803, 804, and 806). Observed and calculated waveforms are shown as black and red lines, respectively. The horizontal axis indicates the elapsed time from the centroid moment time of the mainshock. The horizontal gray bars indicate the time range of the data used in the inversion analysis in *Saito et al.* [2011]. The nonlinear dispersive (NDS) tsunami equations were numerically solved for the calculated waveforms.

based on different tsunami equations with a high-resolution source model and used the observed records as references to reliably examine the propagation process. Section 2 explains our simulation settings. We used the nonlinear dispersive (NDS), the linear dispersive (DSP), and the nonlinear long wave (NLL) tsunami equations. A detailed explanation of the simulation scheme for the NDS equations is provided in Appendix A. In Section 3, the simulation results for the 2011 Tohoku-Oki earthquake tsunami are shown and compared with the actually observed waveforms and inundation area. We illustrate how the dispersion and the nonlinear effects appeared in the records actually observed for the Tohoku tsunami. We also discuss the details of the dispersion, the nonlinearity, and the energy dissipation near the coast. In Section 4, our conclusions are presented.

2. Tsunami Simulation

The 2011 Tohoku-Oki earthquake occurred off the shore of northeastern-Honshu, Japan, and the tsunami excited by the earthquake propagated across the Pacific Ocean. The range of the tsunami simulation is shown in Figure 1. The bathymetry data used in the simulation were compiled from 1350 m, 450 m, 150 m, and 50 m grid resolution bathymetry data provided by Central Disaster Management Council, Government of Japan. The area is represented by 3200 $\times$ 2600 grids with a grid spacing of 500 m. The initial tsunami height distribution was used, which was estimated by the waveform inversion analysis in *Saito et al.* [2011]. This source model was characterized by a steep bump higher than 8 m near the trench and

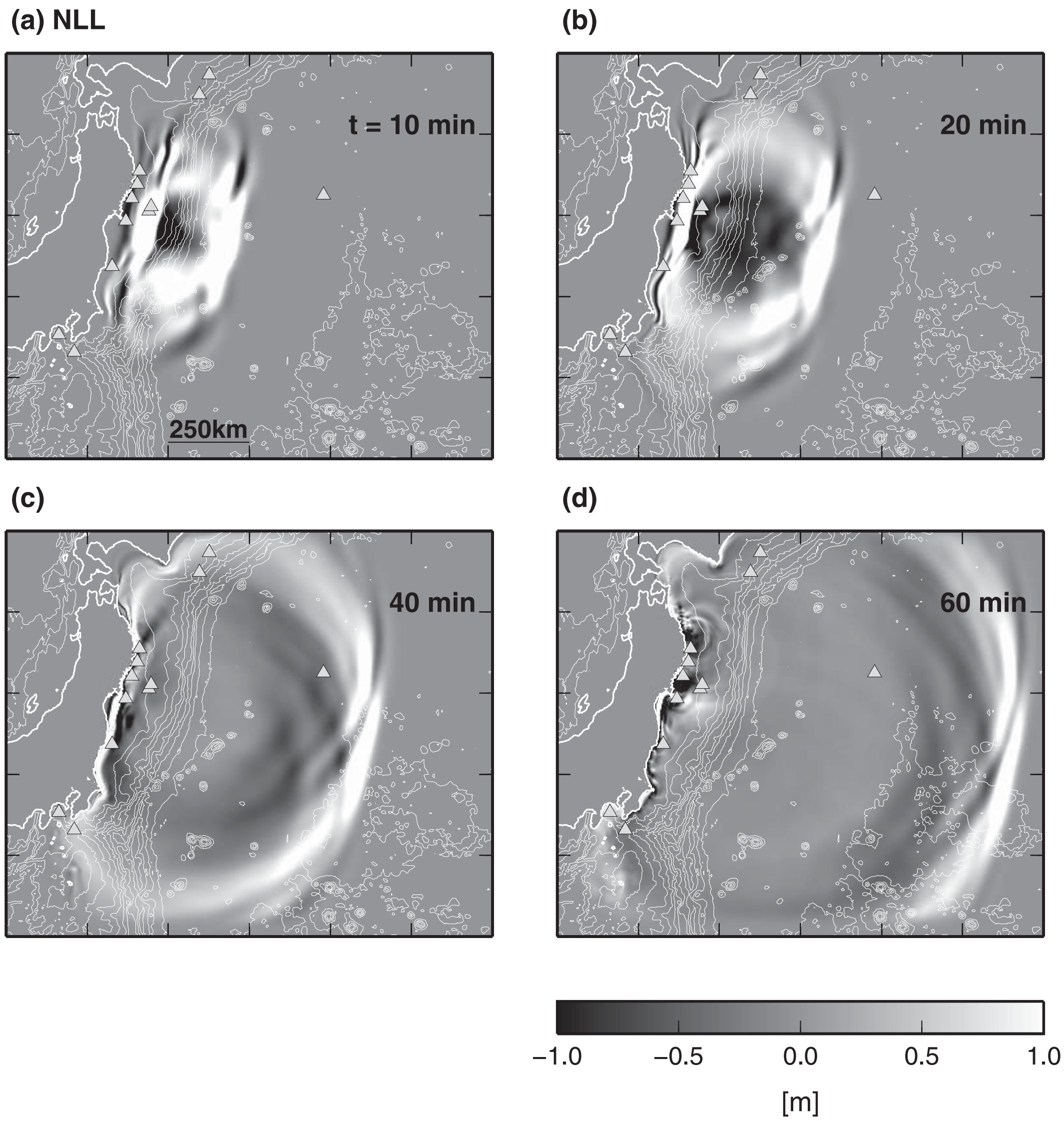

Figure 4. Simulation results of the nonlinear long-wave (NLL) tsunami equations. Tsunami height distribution at the elapsed time of (a) 10, (b) 20, (c) 40, and (d) 60 min from the simulation start time.

characterized by rich short-wavelength components. *MacInnes et al*. [2013] reported that the source model of *Saito et al*. [2011] reproduced the tsunami waveforms recorded at the deep sea better than other models.

This study employs the nonlinear dispersive tsunami equations (NDS) with the *x-y* coordinates in the horizontal space [e.g., *Iwase et al*., 1998]:

$$\frac{\partial \eta}{\partial t}+\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}=0 \tag{1}$$

$$\begin{gathered}\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+g_0\frac{\partial \eta}{\partial y}= \\ \frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}\right]-\frac{g_0 n^2}{(h+\eta)^{\frac{1}{3}}}\frac{u\sqrt{u^2+v^2}}{h+\eta}\end{gathered} \tag{2}$$

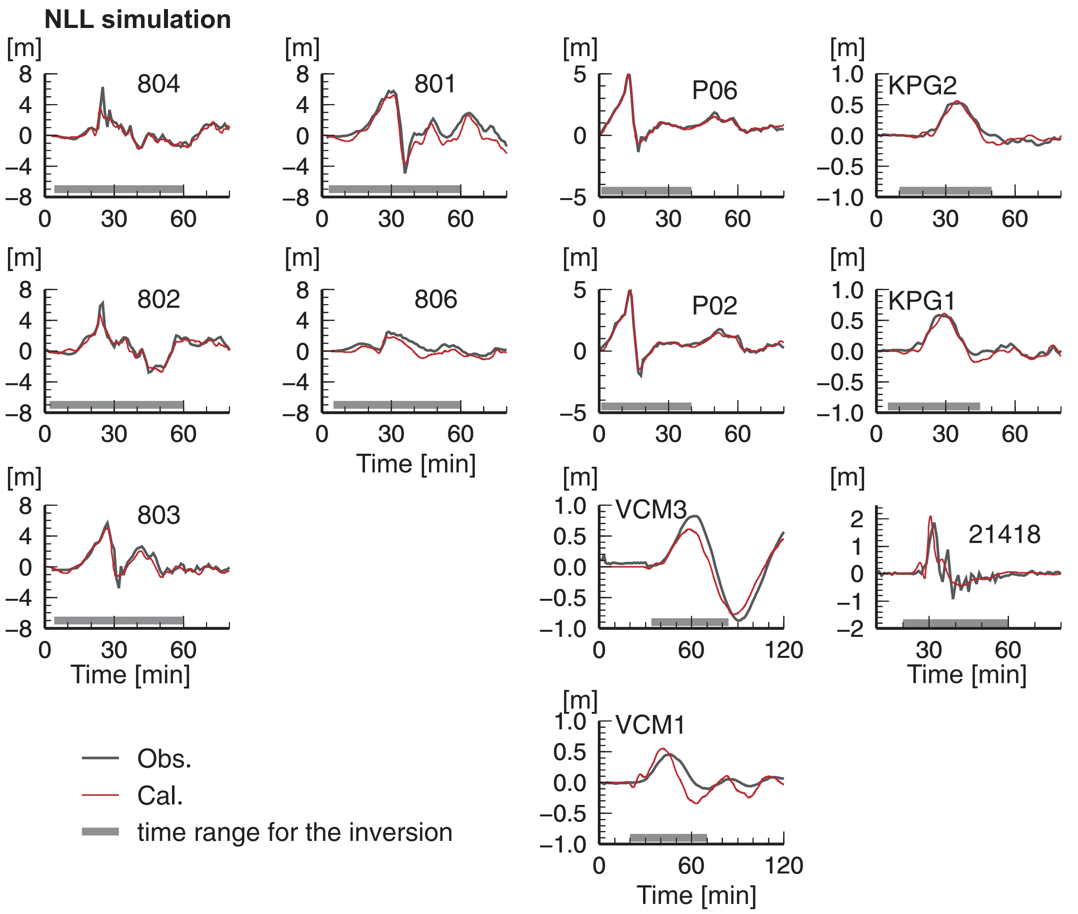

Figure 5. Tsunami waveforms recorded at the stations with the ocean-bottom pressure gauges (KPG1, KPG2, VCM1,VCM3, P02, P06, and 21418) and GPS wave gauges (801, 802, 803, 804, and 806). Observed and calculated waveforms are shown as black and red lines, respectively. The horizontal axis indicates the elapsed time from the centroid moment time of the mainshock. The horizontal gray bars indicate the time range of the data used in the inversion analysis in *Saito et al.* [2011]. The nonlinear dispersive (NLL) tsunami equations were numerically solved for the calculated waveforms.

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+g_0\frac{\partial \eta}{\partial y}=\frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}\right]-\frac{g_0 n^2}{(h+\eta)^{\frac{1}{3}}}\frac{v\sqrt{u^2+v^2}}{h+\eta} \tag{3}$$

where the parameters u and v are the vertically averaged horizontal velocity components, η is the water height from the sea surface at rest, h is the water depth, and g_0 is the gravitational constant. The parameter n is Manning's coefficient and we set $n=0.03$ ($m^{-1/3}$ s) [e.g., *Satake*, 1995]. Inundation is represented by a moving boundary condition between dry and wet cells [e.g., *Iwasaki and Mano*, 1979]. We numerically solve these equations by a finite difference scheme and its detailed procedure is explained in Appendix A.

Other sets of the tsunami equations were also used for comparison. By neglecting the nonlinear terms in Eqs. (1)–(3) assuming that the tsunami height is much smaller than the sea depth ($\eta << h$), the linear dispersive equations (DSP) were obtained as follows [e.g., *Saito et al.*, 2010]:

$$\frac{\partial \eta}{\partial t}+\frac{\partial}{\partial x}\{hu\}+\frac{\partial}{\partial y}\{hv\}=0 \tag{4}$$

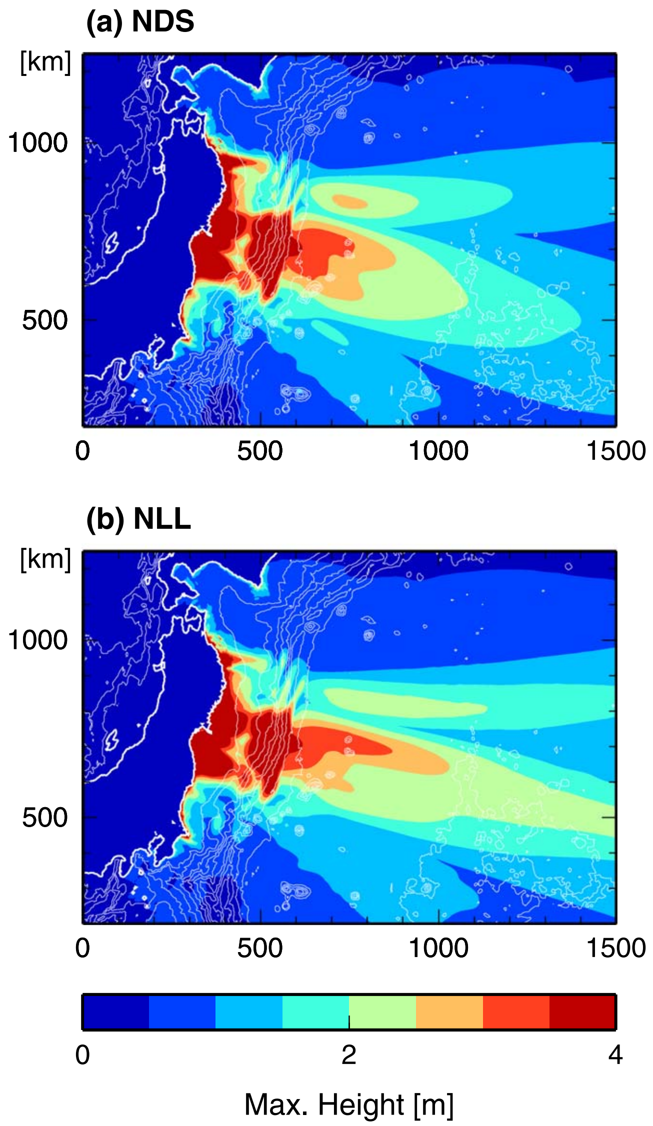

Figure 6. The maximum tsunami height distribution calculated from the initial tsunami height distribution shown in Figure 1 based on (a) the nonlinear dispersive tsunami (NDS) equations and (b) the nonlinear long-wave (NLL) equations.

$$\frac{\partial u}{\partial t}+g_0\frac{\partial \eta}{\partial x}=\frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\{hu\}+\frac{\partial}{\partial y}\{hv\}\right] \tag{5}$$

$$\frac{\partial v}{\partial t}+g_0\frac{\partial \eta}{\partial y}=\frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\{hu\}+\frac{\partial}{\partial y}\{hv\}\right]. \tag{6}$$

By neglecting the dispersion terms in Eqs. (1)–(3) assuming that wavelength is much larger than the sea depth ($\lambda >> h$), the nonlinear long wave equations (NLL) were obtained as follows:

$$\frac{\partial \eta}{\partial t}+\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}=0 \tag{7}$$

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+g_0\frac{\partial \eta}{\partial x}=-\frac{g_0 n^2}{(h+\eta)^{\frac{1}{3}}}\frac{u\sqrt{u^2+v^2}}{h+\eta} \tag{8}$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+g_0\frac{\partial \eta}{\partial y}=-\frac{g_0 n^2}{(h+\eta)^{\frac{1}{3}}}\frac{v\sqrt{u^2+v^2}}{h+\eta}. \tag{9}$$

By comparing the results of these sets of tsunami equations, we investigated the contributions from the dispersion and nonlinear effects.

3. Results and Discussion

3.1. Results of Offshore Tsunami

Figures 2a–2d are the results of the NDS simulation (Eqs. (1) – (3)), which show the tsunami height distribution at the elapsed times of 10, 20, 40, and 60 min (Movie S1, supporting information). The tsunami propagates with high velocity ($\sim$200 m/s) in the deep ocean ($\sim$4 km sea depth). The dispersive tsunami which shows that long-wavelength tsunami propagates faster than short-wavelength tsunami is clearly observed in the Pacific Ocean (Figures 2c and 2d). Furthermore, the tsunami propagates with low velocity in the shallow water near the coast and arrives at the coast of northeastern Japan at 40 min. The tsunami energy is trapped near the coast and the sea surface continues to oscillate after the tsunami arrives (Figure 2d).

Figure 3 shows the calculated tsunami waveforms (red line) together with the waveforms actually observed at the Tohoku tsunami (black line). The station locations are plotted in Figure 1 as triangles.

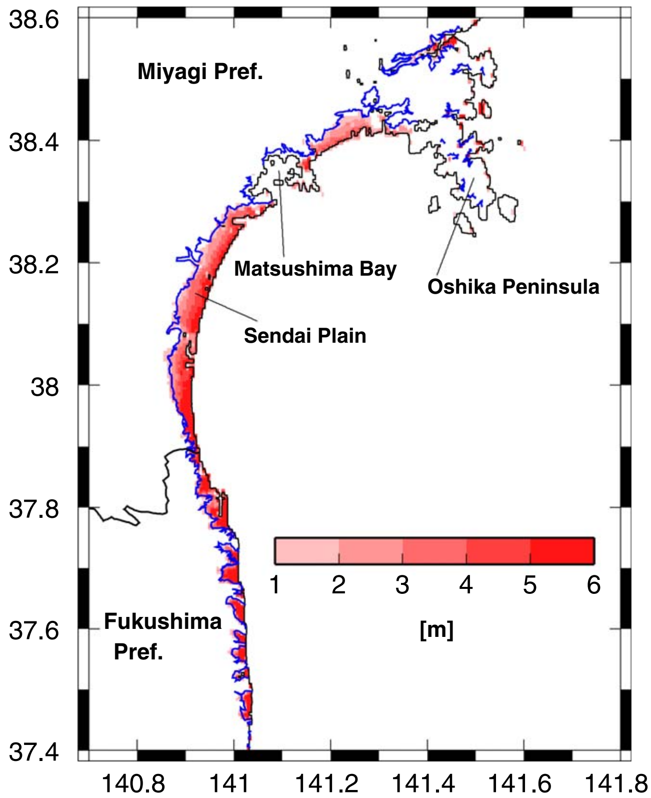

Figure 7. The inundation height in and around the Sendai Plain, which was calculated based on the nonlinear dispersive tsunami (NDS) equations. The inundation limit read from the aerial photos [*Nakajima and Koarai*, 2011] is shown as blue line.

Stations 801, 802, 803, 804, and 806 are the GPS wave gauges operated by the Port and Airport Research Institute (PARI). Stations P02 and P06 are the pressure gauges, which are temporal stations that were installed and operated between June 2010 and May 2011 by Tohoku University [*Hino et al*., 2009]. Stations VCM1 and VCM3 are the pressure gauges operated by the National Research Institute for Earth Science and Disaster Prevention (NIED) [*Eguchi et al*., 1998]. Stations KPG1 and KPG2 are the pressure gauges operated by the Japan Agency for Marine Earth Science and Technology (JAMSTEC) [*Hirata et al*., 2002]. Station 21418 is the pressure gauge of the DART system [*Titov et al*., 2005]. To retrieve the tsunami signals from the original records we apply a low-pass filter with a corner frequency of 2 min for the removal of short period wind and seismic noise. We then approximate the tidal component by fitting a polynomial and remove the tides from the record. Station 21418 recorded the largest tsunami height ($\sim$ 2m) among the DART stations for this event. The waveforms calculated by the NDS simulation successfully reproduced the observed tsunami waveforms. This supports the validity of the source model and the tsunami propagation simulations employed in this study.

We then compared the results of the NDS simulations with those of the NLL simulation (equations (7)–(9)) in order to examine the contributions from the wave dispersion. Figures 4a–4d show the results of the NLL simulation corresponding to the NDS simulation results of Figures 2a–2d (Movie S2, supporting information). In the results of the NLL simulations, the dispersive tsunami does not appear in the deep ocean, but shows a solitary wave propagation (Figures 4c and 4d). This is due to an over simplification of the NLL simulations. The NLL equations neglect the dispersive term (the first term on the right hand side in Eqs. (2) and (3)) due to the assumption that the tsunami wavelength is much longer than the sea depth ($\lambda >> h$). However, in the Tohoku tsunami, the tsunami wavelength was not much longer than the sea depth ($h \sim 4$ km) due to the steep bump near the trench ($\lambda \sim 20$ km), which was composed of rich short-wavelength components (see Figure 1). Figure 5 shows the tsunami waveforms calculated by the NLL simulation with the observed waveforms. The NLL simulation reproduced the observed records very well for almost all of the stations except for station 21418. The comparison of Figures 3 and 5 reveals that the oscillatory wave observed at the station 21418 is caused by the wave dispersion.

The maximum amplitude distribution across the ocean provides us with important information that illustrates the nature of the tsunami. We examined the contribution of the wave dispersion to the maximum amplitude distribution. Figures 6a and 6b compare the maximum amplitude distributions obtained from the NDS simulation and the NLL simulation, respectively. The range for the maximum amplitude that is larger than 2 m extends to more than 1000 km for the NLL simulation (Figure 6b),

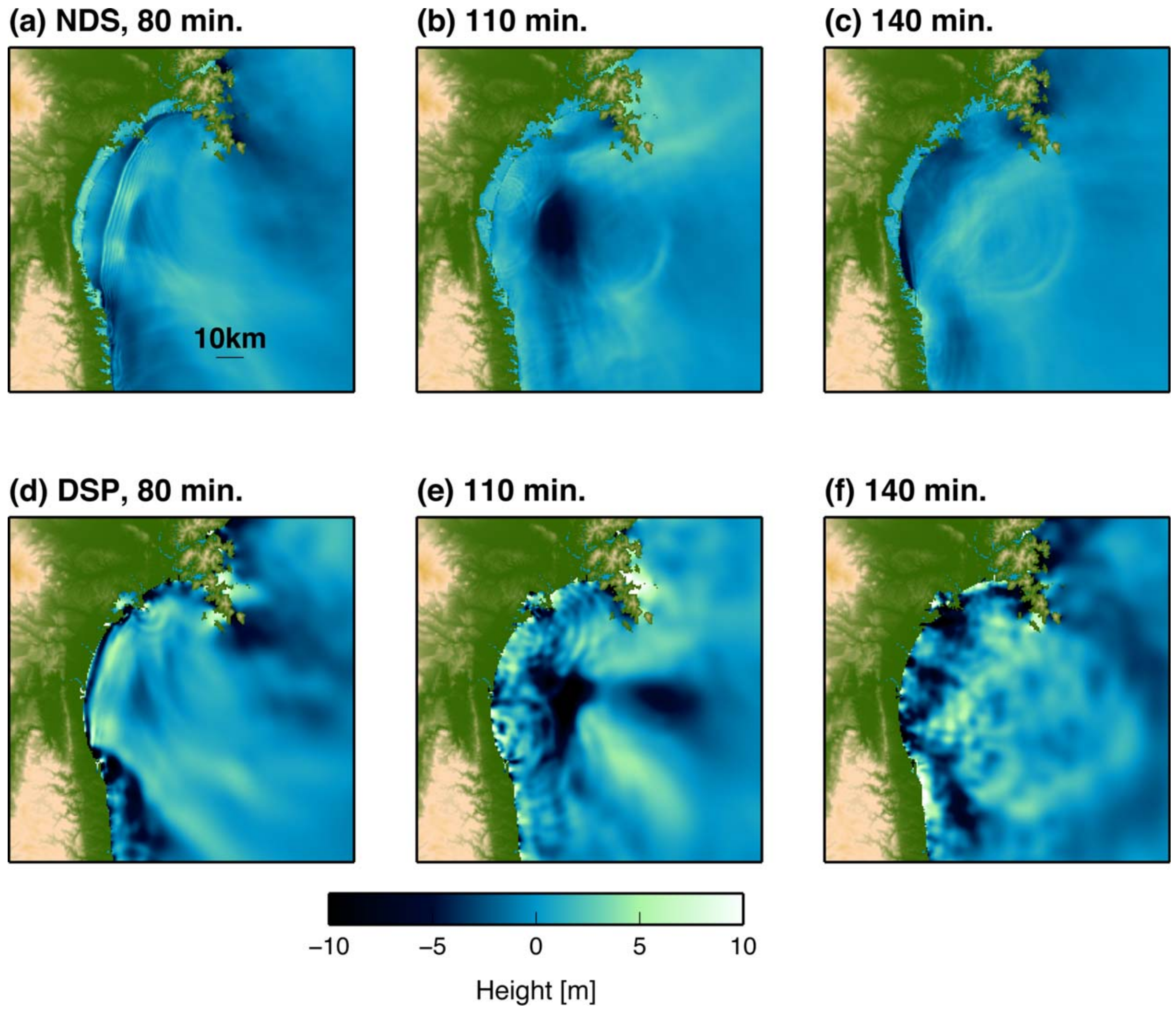

Figure 8. The snapshots of the tsunami propagation near the coast of the Sendai Plain. The snapshots calculated based on the nonlinear dispersive (NDS) equations at the elapsed time of (a) 80 min, (b) 110 min, and (c) 140 min. The snapshots calculated based on the linear dispersive (DSP) equations at the elapsed time of (d) 80 min, (e) 110 min, and (f) 140 min.

while it extends only to 600 km if we correctly include the wave dispersion in the NDS simulation (Figure 6a). The wave dispersion becomes more dominant as the wavelength becomes smaller [e.g., *Stoker*, 1958]. Hence, the wave dispersion is more dominant in the direction perpendicular to the fault strike than in any other direction, since the tsunami wavelength is shorter in the direction perpendicular to the fault strike [*Saito et al*., 2010, Figure B1]. It should be noted that the highest tsunami appears in the direction perpendicular to the fault strike due to the tsunami source radiation pattern [e.g., *Kajiura*, 1970], which makes it important to take the wave dispersion into account for the maximum amplitude distribution in the deep ocean.

3.2. Discussion on the Wave Dispersion

Dispersion occurs when a tsunami propagates across a deep ocean. It is important to include the dispersion for precise waveform modeling. Some studies reported that the wave dispersion (or physical dispersion) could be modeled by creating a numerical dispersion by setting the grid spacing artificially wide enough to create the numerical dispersion. This method works well for rapid tsunami prediction purposes [e.g., *Tang et al*., 2012; *Zhou et al*., 2012]. However, this method is inappropriate when attempting to reproduce the detailed features of the tsunami waveform, because the grid spacing for the numerical dispersion is too

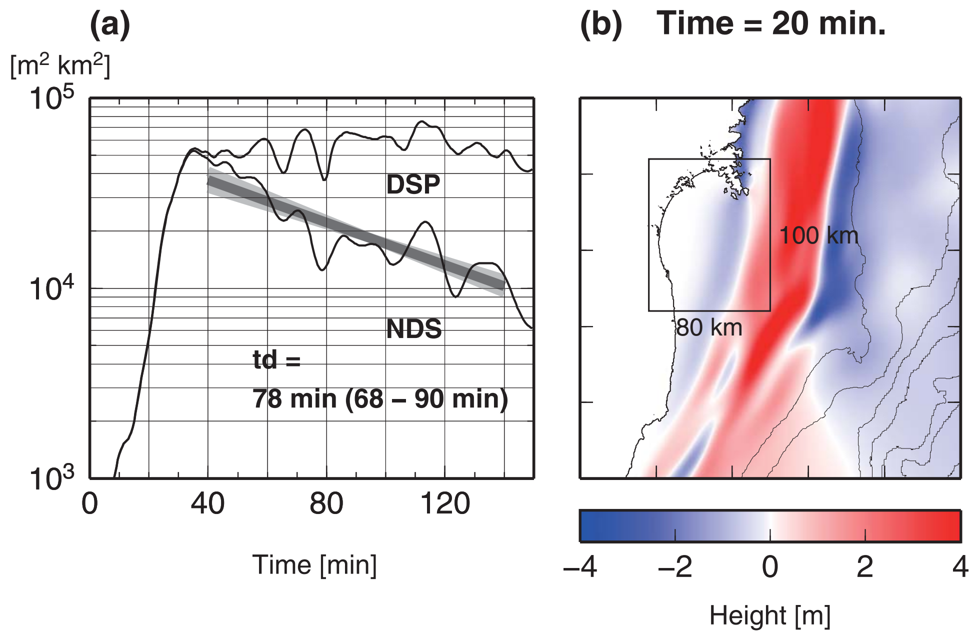

Figure 9. (a) The temporal change of the tsunami energy ($\int\int_S |\eta|^2 dS$) around the Sendai Plain as calculated from the nonlinear dispersive tsunami (NDS) simulation and the linear dispersive tsunami (DSP) simulation. The regression line $E_0 exp[-t/t_d]$ was fitted for the energy with an elapsed time of 40–140 min. The decay time t_d was estimated as $t_d = 78$ min, which ranged from 68 to 90 min when the deviation of the data from the regression line was taken into consideration. (b) The calculated tsunami height distribution at the elapsed time of 20 min by the NDS simulation. The area for the energy estimation in Figure 9a is marked with a rectangle.

coarse to represent the short wavelength components of the tsunami source or the initial tsunami-height distribution. An appropriate grid spacing of 2 min for the wave dispersion in the Pacific Ocean is too coarse to fully reproduce the tsunami source of the Tohoku-Oki earthquake, which had rich short wavelength components and a steep bump near the trench.

Our simulations identified a clear dispersive wave in the tsunami waveform actually recorded at the station 21418. This result is not the same as the result of *Kirby et al*. [2012], in which the oscillatory dispersive tail did not appear in the simulation results of the station 21418. This difference would be due to the difference of the tsunami source model used in the tsunami simulations. We will discuss the effect of the tsunami source on the wave propagation in 3.6. For the 2011 Tohoku-Oki earthquake tsunami, it would be significant to reproduce the tsunami waveform recorded at station 21418, because the record that is located near the source (~500 km) can provide a vast amount of information on the source which is free from distortion caused by the path and site effects. It also shows a clear dispersive wave which can be effectively used for extracting a vast amount of information about the short wavelength component of the source [e.g., *Saito et al*., 2011; *Grilli et al*., 2012].

3.3. Results of Inundation and Near-Shore Tsunami

Figure 7 shows the observed and calculated inundation area around the Sendai Plain, which was the widest inundation area for this event. The solid blue lines indicate the observed inundation limit that was read from the results of *Nakajima and Koarai* [2011]. The observed inundation limit is ~4 km from the coast in the Sendai Plain. Since it was difficult to recognize the difference between the inundation areas calculated by the NDS equations and the NLL equations by eye, only the simulation result of the NDS equations is indicated in Figure 7. The simulation result indicates that the inundation depth that is deeper than 1 m corresponds well with the observed inundation limit.

Figure 8 compares the tsunami propagation around the Sendai Plain at the elapsed times of 80, 110, and 140 min, which were calculated based on the NDS equations (Figures 8a–8c) (Movie S3,

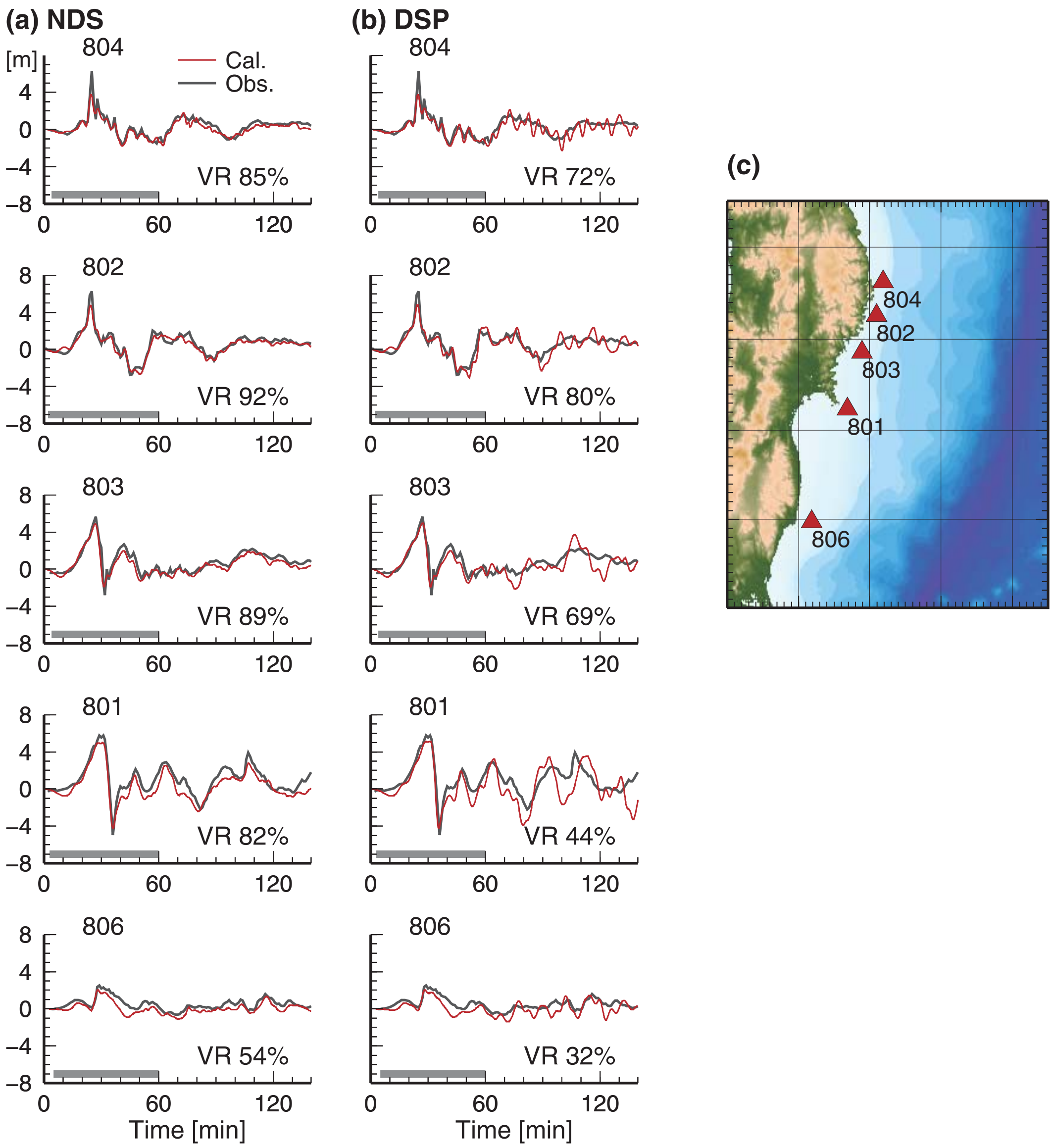

Figure 10. Comparisons of the calculated (red) and observed (black) tsunami waveform for the station with GPS tsunami gauges (804, 802, 803, 801, and 806). (a) The nonlinear dispersive equations (NDS) and (b) linear dispersive equations (DSP) were used for the calculation. The variance reduction (VR) for each calculated waveform is listed in each bin. (c) The locations of the GPS tsunami gauges.

supporting information) and the DSP equations (Figures 8d–8f) (Movie S4, supporting information). At the elapse time of 80 min, a negative-amplitude wave arrives at the Sendai coast in Figure 10d and Movie S4. The corresponding wave in the NDS simulation propagates slower than that of the DSP simulation because the negative amplitude tsunami propagates slower when the nonlinear equations are used (Figure 8a and Movie S3). The NDS simulation includes the inundation process, whereas the DSP simulation does not. Figures 8 display a significant difference between the simulation results of the waves reflected from the coast. The NDS simulation results show that the tsunami inundated the coast and the amplitude of the waves reflected from the coast becomes small (Figures 8a–8c). This suggests that the tsunami energy dissipates significantly due to the bottom friction as the tsunami propagates near the coast. Conversely, the DSP equations (equations (4)–(6)) do not include the nonlinear process and preserve the total energy. Thus, the DSP simulation results show a tsunami with a larger amplitude that was reflected from the coast without the energy dissipation (Figures 8d–8f).

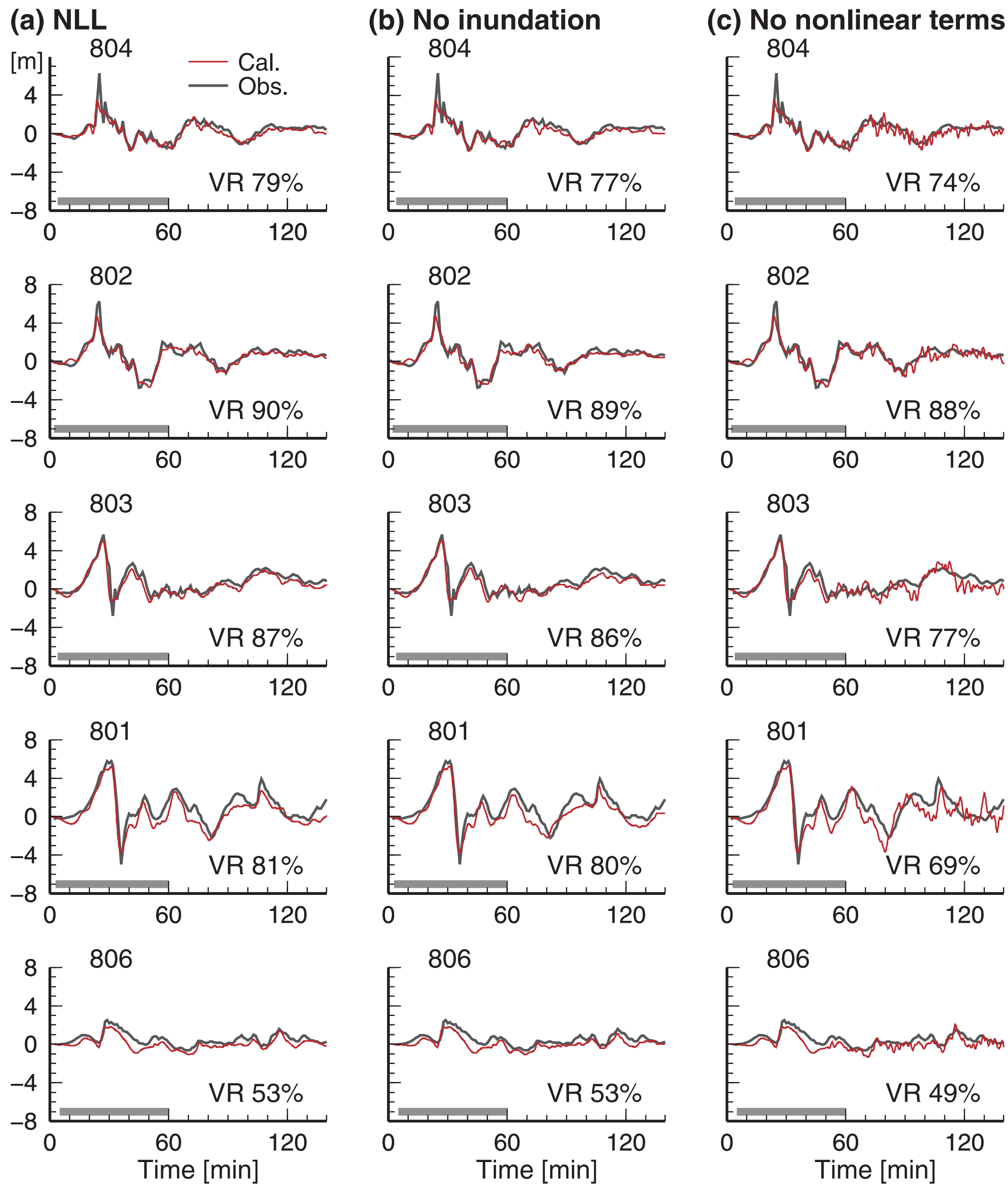

Figure 11. Comparisons of the calculated (red) and observed (black) tsunami waveforms at the stations with GPS tsunami gauges (804, 802, 803, 801, and 806). (a) The nonlinear long-wave equations (NLL), (b) the NLL equations with the nonlinear terms and without the inundation effect, and (c) the NLL equations with the inundation effect without the nonlinear terms. The variance reduction (VR) for each calculated waveform is listed in each bin.

We then attempted to quantify the energy dissipation by evaluating the temporal change of the energy near the coast. The tsunami energy per unit area is given by $\rho_0 g_0 |\eta|^2$, where ρ_0 is the water density, g_0 is the gravitational constant, and η is the sea surface variation [e.g., *Pedlosky*, 2003], which is proportional to the square of the waveform $|\eta|^2$ and independent of the sea depth. Assuming that ρ_0 and g_0 are constant in space, we calculate the tsunami energy $\int\int_S |\eta|^2 dS$ trapped in the Bay of Sendai. The area S of this integration is shown as a solid line in Figure 9b. Figure 9a plots the temporal change of the energy in this rectangular area with respect to the elapsed time from the simulation start time. The NDS simulation results indicate that the tsunami energy decays almost monotonically with the elapsed time. The energy decay can be fitted by the regression line $E_0 exp[-t/t_d]$, where t is the elapsed time and the decay time t_d was estimated as

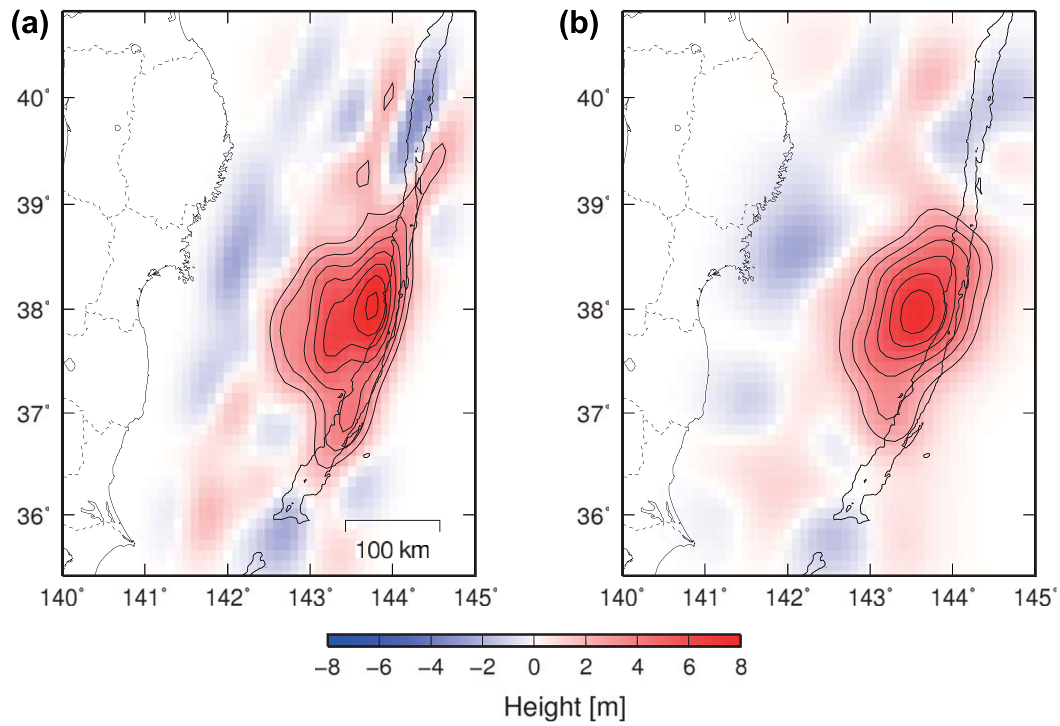

Figure 12. Initial tsunami height distributions for the tsunami simulations [*Saito et al.* 2011]: (a) a high-resolution model, and (b) a low-resolution model. The initial tsunami height distribution is contoured from 2 to 8 m at intervals of 1 m.

78 min. Contrary to the case of the NDS simulation, if we calculate the temporal change of the energy based on the result of the DSP simulations, the energy does not show a decay with the elapsed time, indicating that the energy does not dissipate in this area.

In order to prove the tsunami energy dissipation by the observational facts, we compare the observed and calculated waveforms in Figure 10. Figure 10a shows the observed tsunami waveforms (black lines) and the waveforms calculated by the NDS simulation (red lines). Gray bars indicate the length of the time window employed in the estimation of the tsunami source [*Saito et al.*, 2011]. The NDS simulation reproduces the tsunami waveforms surprisingly well including the records that were not used for the source estimation. In contrast, when we used the same tsunami source in the DSP equations (Figure 10b), the later part of the waveforms differs from that of the observed waveforms; the calculated waveform overestimated the amplitude. If we check the fitting with the variance reduction (VR), the NDS simulation shows VR values that are significantly larger than those of the DSP simulations for all 5 stations. This suggests that our NDS simulation can properly include the energy dissipation near the coast.

3.4. Dissipation Mechanisms: Inundation and Nonlinear Terms

For the Tohoku tsunami, many studies considered the nonlinear process and the inundation to successfully simulate the tsunami waveforms [e.g., *Tang et al.*, 2012; *Wei et al.*, 2013; *Yamazaki et al.*, 2013]. However, the most dominant factor contributing to the tsunami waveforms that were observed offshore was not obvious. This subsection investigates the mechanism of the energy dissipation by comparing the simulation results of various equations to determine the most dominant effect on the tsunami waveform that is observed offshore (Figure 10). Figure 11 compares the tsunami waveforms by different simulation settings. Figure 11a shows the waveforms of the NLL simulation, which properly reproduce the observed waveforms as well as the waveforms of the NDS simulation shown in Figure 10a. If we check the fitting with the VR, the NDS simulation shows slightly larger VR values for all 5 stations. This indicates that the nonlinear effects mainly contribute to the wave synthesis for the station, although the dispersion slightly improves the reproduction even near the

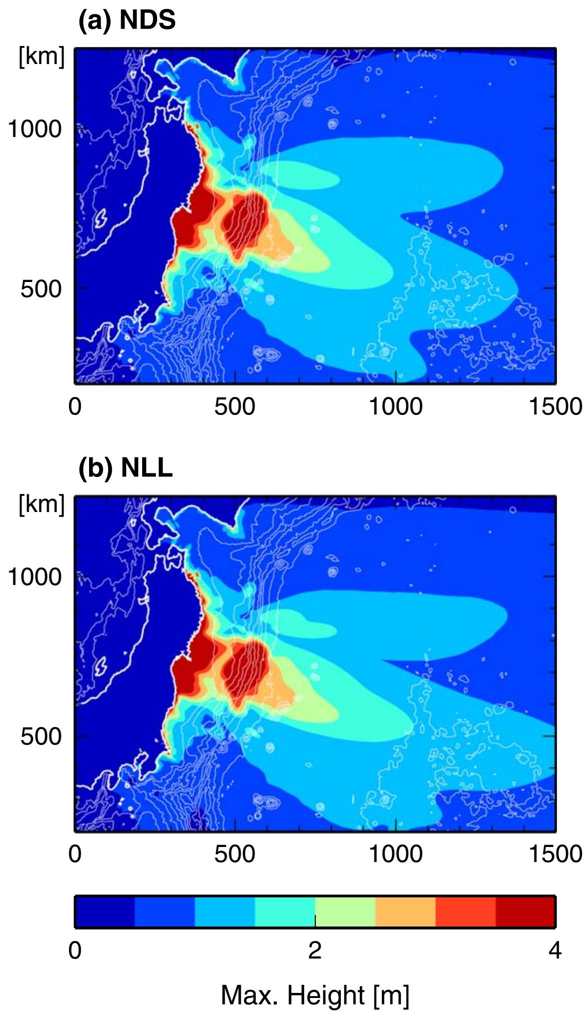

Figure 13. The maximum tsunami height distribution calculated from the low-resolution initial tsunami height distribution (Figure 12b) based on (a) the nonlinear dispersive tsunami (NDS) equations and (b) the nonlinear long-wave (NLL) equations.

coast. Figure 11b shows the waveforms calculated by the NLL simulation without the inundation effect. Figure 11c shows the waveforms produced by the NLL simulation with the inundation and without the nonlinear terms of the advection and bottom friction (i.e., by the linear long-wave equations with the inundation effect). The waveforms of the simulation without the inundation (Figure 11b) show almost the same waveform as those of the NDS and the NLL simulations (Figures 10a and 11a). In contrast, the waveforms of the simulation without the nonlinear terms (Figure 11c) cannot fully reproduce the later tsunami amplitude. This indicates that the nonlinear terms rather than the inundation were intrinsically important for modeling the later-arriving tsunami of the offshore stations for the Tohoku tsunami.

3.5. Discussion on the Nonlinearity

It is widely known that the nonlinear effects of inundations and bottom friction are important for the near-shore areas where the water depth is very shallow ($< \sim 50$ m). For example, by comparing the observed waveforms and synthetic waveforms by varying the value of the bottom friction coefficient, *Satake* [1995] showed how the bottom friction coefficient affects the tsunami waveform and concluded the importance of the nonlinear effects on waveform modeling. However, he showed only a single waveform and the agreement between the calculation and the observation was not good. In this study, a better agreement between the observed and simulated waveforms was achieved (Figure 10). This showed the importance of the nonlinear effects. This good agreement attributed to the use of high quality tsunami waveforms that were recorded offshore, reliable topography data, and a source model. Many high-quality data were recorded for the Tohoku earthquake due to the development of instrumental technology and the deployment of many stations in the offshore regions. This differed compared to the recordings of other previous earthquake tsunamis.

The tsunami decay process has attracted much interest among tsunami researchers for many years. *Munk* [1963] constructed a theoretical model of the decay process by phenomenologically including the energy dissipation at the reflection of the coasts, which pioneered the studies of the decay process. *Van Dorn* [1984, 1987] investigated the decay process and obtained a decay time of ~ 22 hr for the Pacific Ocean. Figure 9 shows that the early stage of the tsunami energy ($<$ elapsed time of 150 min) is characterized by a decay time t_d of 78 min. In contrast, the energy decay at a much later stage (elapsed time of $\sim$48 hr) is characterized by a much longer decay time ($t_d \sim 24$ h) [see *Saito et al.* 2013, Figure 11; *Tang et al.*, 2012]. This value is almost the same as that obtained for the Pacific Ocean by *Van Dorn* [1984]. Considering that the tsunami amplitude largely varied from 10^0 m to 10^{-3} m from the beginning to the end of the Tohoku tsunami in the Pacific Ocean [e.g.,

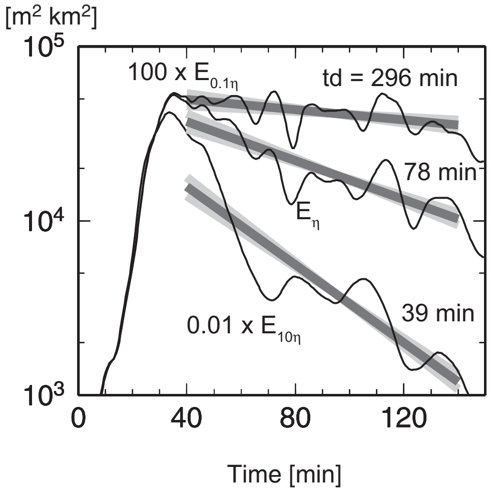

Figure 14. The temporal change of the tsunami energy around the Sendai Plain as calculated from the nonlinear dispersive tsunami (NDS) simulation. A large-amplitude source was artificially created by multiplying the number 10 to the initial tsunami height distribution shown in Figure 1. The energy decay for the large-amplitude source is given by $E_{10\eta}$ and was plotted by multiplying it by 0.01 ($=0.1^2$). A small-amplitude source was artificially created by multiplying 0.1 to the initial tsunami height distribution shown in Figure 1. The energy decay for the large-amplitude source is given by $E_{0.1\eta}$ and was plotted by multiplying it by 100 ($=10^2$).

Fine et al., 2012; *Saito et al.*, 2012], the nonlinear nature of the attenuation is considered to have played an important role in causing a significant difference in the tsunami energy attenuation during the early stage (elapsed time of ~80 min) and late stage (elapsed time of ~48 hr) of the Tohoku tsunami decay process. We investigate a nonlinear nature (an amplitude dependence) of tsunami decay also in 3.6.

3.6. Source Effects on the Propagation Process

We have shown that appropriate tsunami equations are necessary to precisely model the propagation process. We now show that the source properties also strongly control the propagation process. The higher-resolution source model and the lower-resolution source model were used for a comparative simulation (Figure 12). The higher-resolution model is the same as the model shown in Figure 1. The lower-resolution source model is the same as the model shown in Figure 3b in *Saito et al*. [2011]. The two source models were estimated by using the same method with the same data set, but the lower-resolution source model has poor short-wavelength components because it is represented by fewer numbers and larger source elements. The lower-resolution model was able to reproduce all of the records except for the dispersive wave at station 21418 [*Saito et al*., 2011, Figure S1b]. Figure 13 shows the maximum amplitude distributions calculated from the lower-resolution source model. If the lower-

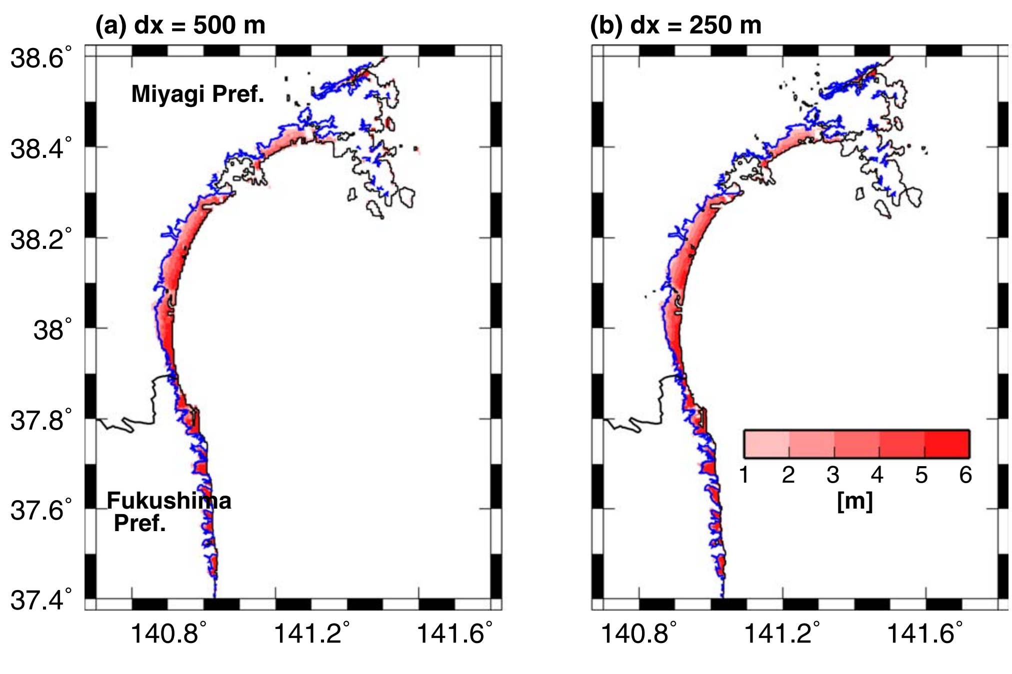

Figure 15. The inundation height in and around the Sendai Plain. (a) The nonlinear long-wave equations (NLL) were numerically solved by using a grid spacing of 500 m and (b) the NLL equations with a grid spacing of 250 m.

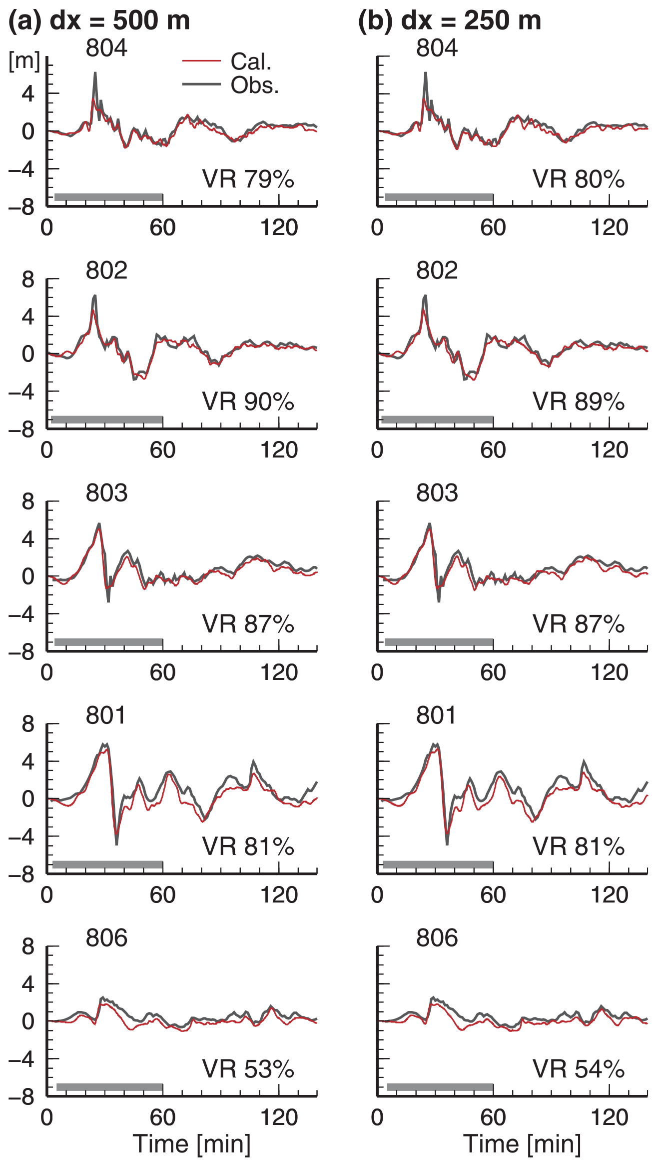

Figure 16. Comparisons of the calculated (red) and observed (black) tsunami waveforms at the stations with GPS tsunami gauges (804, 802, 803, 801, and 806). (a) The nonlinear long-wave equations (NLL) were numerically solved by using a grid spacing of 500 m and (b) the NLL equations with a grid spacing of 250 m. The variance reduction (VR) for each calculated waveform is listed in each bin.

resolution source model was used, the difference between the results of the NDS and the NLL simulations becomes smaller than that for the case with the higher-resolution source model (Figure 6). This is because the dispersion dominates more in short wavelength tsunamis, so that the dispersion became weak in the lower-resolution source model characterized by poor short wavelength components.

Since the nonlinear effects depend on the wave amplitude, it would also be interesting to show how the energy dissipation could change if the amplitude of the tsunami was larger or smaller. Figure 14 shows that the energy dissipation for different source amplitudes. A large-amplitude source was artificially created by multiplying the number 10 to the original initial tsunami height distribution. The energy decay for the large-amplitude source is given by $E_{10\eta}$ and plotted in the figure where we multiplied 0.01 ($=0.1^2$) to the value of $E_{10\eta}$ to compare it with the original energy decay E_η. Similarly, a small-amplitude source was created by multiplying 0.1 to the original initial tsunami height distribution and the energy decay that is plotted for the small-amplitude source is given by $100 \times E_{0.1\eta}$. The energy decay of the large-amplitude source $E_{10\eta}$ shows a strong attenuation characterized by a short decay time of $t_d = 39$ min (the estimation of t_d can range from 36 to 43 min), while the energy decay of the small-amplitude source $E_{0.1\eta}$ indicates a weak attenuation characterized by a long decay time of $t_d = 296$ min (the estimation can range from 207 to 518 min). These are considered to be reasonable results. As the tsunami height becomes higher, the velocity increases more and the bottom friction becomes larger. Therefore, the tsunami energy dissipates more as the tsunami height increases, as shown in Figure 14.

The above two examples confirm that not only the tsunami equations but also the source model can significantly affect the propagation process of the wave dispersion and nonlinearity. Therefore, in order to correctly examine the contribution of the wave dispersion and nonlinear effects in the Tohoku tsunami, an appropriate source model of the Tohoku tsunami is needed.

3.7. Discussion on Tsunami Waveform Modeling

This study, which uses the NDS equations and a high-resolution source model, has reproduced the tsunami waveforms for the near-shore and offshore regions. The NDS equations include both the nonlinear and dispersive effects. However, solving the NDS equations has a higher computational cost than those for the DSP and NLL equations. The NDS simulation requires grid spacing that is finer than that of the DSP because of the near-coast modeling. The NDS simulation also needs to employ an implicit scheme to stably calculate the dispersion while the NLL employs an explicit scheme. If you consider a practical application for tsunami early warning, developing a more efficient method to solve the NDS equations on supercomputers is necessary [e.g., *Lynett et al*., 2002; *Saito and Furumura*, 2009; *Kirby et al*., 2012; *Shi et al*., 2012; *Oishi et al*., 2013]. For this purpose, it is a good compromise to use the DSP equations for source estimation and propagation across the ocean, and to use the NLL equations for tsunami and inundation predictions near the coast. Note that this conventional method would be useful for real-time forecasts, but it is not suitable for the detailed analysis of a short-wavelength tsunami near the coast and the later tsunami phases reflected from the coasts and observed at offshore regions.

Satake et al. [2013] modeled the tsunami source as a spatially and temporally varying earthquake rupture. The tsunami source used in this study does not have such a time delay process. *Grilli et al*. [2012] suggest a large submarine mass failure occurred northern part of the focal area two minutes after the seismic faulting of the main shock. These can explain why we underestimated the maximum amplitude of the leading wave at station 804. It is desirable to estimate the tsunami source that provides all the details of the tsunami waveforms and runup. To reproduce all the data sets perfectly, it is also important to use more fine scale topography data (e.g., *Baba et al*., 2014], to adjust optimal friction parameter and to examine the difference among friction models [e.g., *Lynett et al*., 2002; *Shi et al*., 2012].

Although the grid spacing used in this study (500 m) may be too coarse to represent small scale topography, the grid spacing was able to correctly reproduce the large-scale inundation area ($> \sim$4 km), such as that in the Sendai Plain, and also properly calculated the reflected wave from the coast as observed offshore (Figures 7 and 10). The NLL simulations results calculated from the grid spacing of 500 m and that from a finer grid spacing of 250 m are shown for the inundation area in Figure 15 and the waveforms in Figure 16. The agreement between the different grid-size simulations validated the use of our simulation with a grid spacing of 500 m and confirmed our conclusions with respect to the inundation area around the Sendai plain and offshore waveforms.

4. Conclusion

We conducted a tsunami simulation of the 2011 Tohoku-Oki earthquake based on the nonlinear dispersive equations by using a source model that was estimated with a short-wavelength tsunami and near-field tsunami inside the focal area. This simulation reproduced the dispersive tsunami recorded at station 21418, which recorded the largest tsunami height ($>$ 2 m) among the DART stations. The dispersion is required to correctly model the propagation in the open ocean, which significantly contributes to the maximum amplitude distribution (Figure 6). The simulation result of the inundation area in the Sendai Plain agreed well with the observation (Figure 7). By including the nonlinear effects of the inundation and the energy dissipation, we correctly simulated the tsunami reflected from the coast (Figure 10). Sets of the simulation that used different tsunami equations indicated that including nonlinear terms rather than the inundation contributes to the precise modeling of the waveform observed at the offshore region (Figure 11). Comparisons of high-quality offshore records and detailed tsunami source/propagation modeling revealed how the wave dispersion and nonlinear effects contributed to the tsunami records observed during the 2011 Tohoku-Oki earthquake.

Appendix A: Finite-Difference Scheme for Nonlinear Dispersive Equations

The finite difference scheme for Eqs. (1)–(3) is described in this appendix. The method used in this study follows the method that was described in *Iwase et al*. [1998]. However, this study uses a vertically averaged horizontal velocity *u* and *v* to avoid numerical instability; the advective terms diverged at the very shallow sea if

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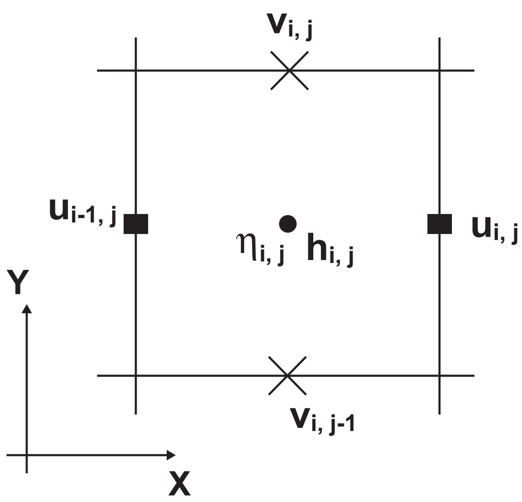

Figure A1. Staggered grids for the finite-difference simulation of nonlinear dispersive equations (equations (1)–(3)).

we used the velocity components integrated along the vertical direction from the sea bottom to the sea surface. A staggered grid was used for the cell located at i on the x axis and at j on the y axis, as shown in Figure A1. The water height η is defined at time $t=n \cdot dt$ and the average horizontal velocity u and v are defined at time $t=(n+1/2) \cdot dt$, where $n=0, 1, 2, \ldots$. The thickness of the water in this cell is given by the sum of the water depth h and the water height η as, $d_{i,j}^n=\eta_{i,j}^n+h_{i,j}^n$. Eq. (1) is expressed in finite difference form as,

$$\eta_{i,j}^{n+1}=\eta_{i,j}^{n}-\frac{\Delta t}{\Delta x}\left\{d_{i+1/2,j}^{n}u_{i,j}^{n+1/2}-d_{i-1/2,j}^{n}u_{i-1,j}^{n+1/2}\right\}-\frac{\Delta t}{\Delta y}\left\{d_{i,j+1/2}^{n}v_{i,j}^{n+1/2}-d_{i,j-1/2}^{n}v_{i,j-1}^{n+1/2}\right\}. \tag{A1}$$

By introducing parameter u^*, we express Eq. (2) in the finite difference form as

$$\frac{u_{i,j}^{n+1/2}-u^*}{\Delta t}+\frac{u^*-u_{i,j}^{n-1/2}}{\Delta t}=\left[-g_0\frac{\partial\eta}{\partial x}-u\frac{\partial u}{\partial x}-v\frac{\partial u}{\partial y}-\frac{g_0 n^2}{(h+\eta)^{1/3}}\frac{u\sqrt{u^2+v^2}}{h+\eta}\right]_{i,j}+\left[\frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial x}\left\{\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}\right\}\right]_{i,j} \tag{A2}$$

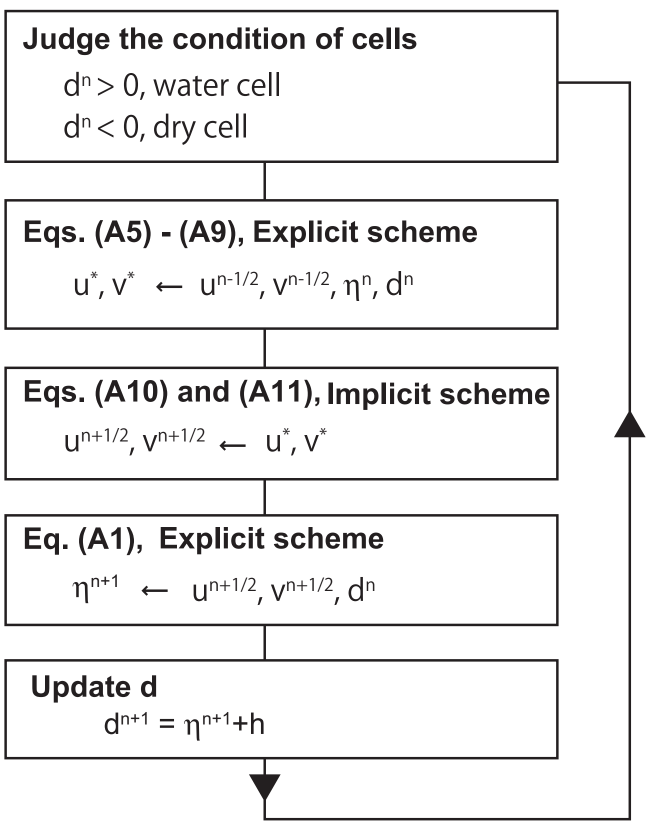

Figure A2. A flowchart for solving the nonlinear dispersive equations.

where the bracket $[\,]_{i,j}$ represents the finite difference form at the ith and jth cell. The finite difference equation (Eq. (A2)) is divided into the following two equations:

$$\frac{u^*-u_{i,j}^{n-1/2}}{\Delta t}=\left[-g_0\frac{\partial\eta}{\partial x}-u\frac{\partial u}{\partial x}-v\frac{\partial u}{\partial y}-\frac{g_0 n^2}{(h+\eta)^{1/3}}\frac{u\sqrt{u^2+v^2}}{h+\eta}\right]_{i,j}, \tag{A3}$$

$$\frac{u_{i,j}^{n+1/2}-u^*}{\Delta t}=\left[\frac{1}{3}h\frac{\partial}{\partial t}\frac{\partial}{\partial x}\left\{\frac{\partial}{\partial x}\{(h+\eta)u\}+\frac{\partial}{\partial y}\{(h+\eta)v\}\right\}\right]_{i,j}. \tag{A4}$$

We explicitly solved Eq. (A3) and implicitly solved Eq. (A4). A similar differentiation was also done for the velocity v of Eq. (3). Eq. (A3) and the corresponding equations for v are written by an explicit scheme as

$$\begin{aligned} u^*_{i,j} = \frac{1}{1+Q^n_{i,j}\Delta t} \Bigg\{ & (1-Q^n_{i,j}\Delta t)u^{n-1/2}_{i,j} - \frac{\Delta t}{\Delta x} g_0 (\eta^n_{i+1,j} - \eta^n_{i,j}) \\ & - \frac{\Delta t}{\Delta x} u^{n-1/2}_{i,j} (\lambda_{11} u^{n-1/2}_{i+1,j} + \lambda_{21} u^{n-1/2}_{i,j} + \lambda_{31} u^{n-1/2}_{i-1,j}) \\ & - \frac{\Delta t}{\Delta y} v^{n-1/2}_{i+1/2,j-1/2} (\mu_{11} u^{n-1/2}_{i,j+1} + \mu_{21} u^{n-1/2}_{i,j} + \mu_{31} u^{n-1/2}_{i,j-1}) \Bigg\}, \end{aligned} \tag{A5}$$

and

$$\begin{aligned} v^*_{i,j} = \frac{1}{1+R^n_{i,j}\Delta t} \Bigg\{ & (1-R^n_{i,j}\Delta t)v^{n-1/2}_{i,j} - \frac{\Delta t}{\Delta y} g_0 (\eta^n_{i,j+1} - \eta^n_{i,j}) \\ & - \frac{\Delta t}{\Delta x} u^{n-1/2}_{i-1/2,j+1/2} (\lambda_{12} v^{n-1/2}_{i+1,j} + \lambda_{22} v^{n-1/2}_{i,j} + \lambda_{32} u^{n-1/2}_{i-1,j}) \\ & - \frac{\Delta t}{\Delta y} v^{n-1/2}_{i,j} (\mu_{12} v^{n-1/2}_{i,j+1} + \mu_{22} v^{n-1/2}_{i,j} + \mu_{32} v^{n-1/2}_{i,j-1}) \Bigg\}, \end{aligned} \tag{A6}$$

where

$$Q^n_{i,j} = \frac{1}{2} \frac{g_0 n_0^2}{(d^n_{i+1/2})^{1/3}} \frac{\sqrt{\left(u^{n-1/2}_{i,j}\right)^2 + \left(v^{n-1/2}_{i+1/2,j-1/2}\right)^2}}{d^n_{i+1/2,j}}, \tag{A7}$$

and

$$R^n_{i,j} = \frac{1}{2} \frac{g_0 n_0^2}{(d^n_{i,j+1/2})^{1/3}} \frac{\sqrt{\left(u^{n-1/2}_{i-1/2,j+1/2}\right)^2 + \left(v^{n-1/2}_{i,j}\right)^2}}{d^n_{i,j+1/2}}. \tag{A8}$$

The parameters λ_{ij} and μ_{ij} in Eqs. (A5) and (A6) are the coefficients of an upwind scheme. The coefficients are chosen as,

$$\begin{aligned} &\lambda_{11}=0, \lambda_{21}=1, \lambda_{31}=-1 \quad \text{for} \quad u^{n-1/2}_{i,j} \geq 0, \\ &\lambda_{11}=1, \lambda_{21}=-1, \lambda_{31}=0 \quad \text{for} \quad u^{n-1/2}_{i,j} < 0, \\ &\mu_{11}=0, \mu_{21}=1, \mu_{31}=-1 \quad \text{for} \quad v^{n-1/2}_{i+1/2,j-1/2} \geq 0, \\ &\mu_{11}=1, \mu_{21}=-1, \mu_{31}=0 \quad \text{for} \quad v^{n-1/2}_{i+1/2,j-1/2} < 0, \\ &\lambda_{12}=0, \lambda_{22}=1, \lambda_{32}=-1 \quad \text{for} \quad u^{n-1/2}_{i-1/2,j+1/2} \geq 0, \\ &\lambda_{12}=1, \lambda_{22}=-1, \lambda_{32}=0 \quad \text{for} \quad u^{n-1/2}_{i-1/2,j+1/2} < 0, \\ &\mu_{12}=0, \mu_{22}=1, \mu_{32}=-1 \quad \text{for} \quad v^{n-1/2}_{i,j} \geq 0, \\ &\mu_{12}=1, \mu_{22}=-1, \mu_{32}=0 \quad \text{for} \quad v^{n-1/2}_{i,j} < 0. \end{aligned} \tag{A9}$$

Eq. (A4) and the corresponding equation for v are written by an implicit scheme as

$$\begin{aligned} & -\frac{h_{i+1/2,j}h_{i+3/2,j}}{3(\Delta x)^2} u^{n+1/2}_{i+1,j} + \left(1 + \frac{2\left(h_{i+1/2,j}\right)^2}{3\Delta x^2}\right) u^{n+1/2}_{i,j} - \frac{h_{i+1/2,j}h_{i-1/2,j}}{3(\Delta x)^2} u^{n+1/2}_{i-1,j} \\ & -\frac{h_{i+1/2,j}h_{i+1,j+1/2}}{3\Delta x\Delta y} v^{n+1/2}_{i+1,j} + \frac{h_{i+1/2,j}h_{i,j+1/2}}{3\Delta x\Delta y} v^{n+1/2}_{i,j} + \frac{h_{i+1/2,j}h_{i+1,j-1/2}}{3\Delta x\Delta y} v^{n+1/2}_{i+1,j-1} - \frac{h_{i+1/2,j}h_{i,j-1/2}}{3\Delta x\Delta y} v^{n+1/2}_{i,j-1} \\ & = -\frac{h_{i+1/2,j}h_{i+3/2,j}}{3(\Delta x)^2} u^*_{i+1,j} + \left(1 + \frac{2\left(h_{i+1/2,j}\right)^2}{3\Delta x^2}\right) u^*_{i,j} - \frac{h_{i+1/2,j}h_{i-1/2,j}}{3(\Delta x)^2} u^*_{i-1,j} \\ & -\frac{h_{i+1/2,j}h_{i+1,j+1/2}}{3\Delta x\Delta y} v^*_{i+1,j} + \frac{h_{i+1/2,j}h_{i,j+1/2}}{3\Delta x\Delta y} v^*_{i,j} + \frac{h_{i+1/2,j}h_{i+1,j-1/2}}{3\Delta x\Delta y} v^*_{i+1,j-1} - \frac{h_{i+1/2,j}h_{i,j-1/2}}{3\Delta x\Delta y} v^*_{i,j-1} \end{aligned} \tag{A10}$$

and,

$$-\frac{h_{i,j+1/2}h_{i,j+3/2}}{3(\Delta y)^2}v_{i,j+1}^{n+1/2}+\left(1+\frac{2\left(h_{i,j+1/2}\right)^2}{3\Delta y^2}\right)v_{i,j}^{n+1/2}-\frac{h_{i,j+1/2}h_{i,j-1/2}}{3(\Delta y)^2}v_{i,j-1}^{n+1/2}$$
$$-\frac{h_{i,j+1/2}h_{i+1/2,j+1}}{3\Delta x\Delta y}u_{i,j+1}^{n+1/2}+\frac{h_{i,j+1/2}h_{i-1/2,j+1}}{3\Delta x\Delta y}u_{i-1,j+1}^{n+1/2}+\frac{h_{i,j+1/2}h_{i+1/2,j}}{3\Delta x\Delta y}u_{i,j}^{n+1/2}-\frac{h_{i,j+1/2}h_{i-1/2,j}}{3\Delta x\Delta y}u_{i-1,j}^{n+1/2}$$
$$=-\frac{h_{i,j+1/2}h_{i,j+3/2}}{3(\Delta y)^2}v_{i,j+1}^{*}+\left(1+\frac{2\left(h_{i,j+1/2}\right)^2}{3\Delta y^2}\right)v_{i,j}^{*}-\frac{h_{i,j+1/2}h_{i,j-1/2}}{3(\Delta y)^2}v_{i,j-1}^{*}$$
$$-\frac{h_{i,j+1/2}h_{i+1/2,j+1}}{3\Delta x\Delta y}u_{i,j+1}^{*}+\frac{h_{i,j+1/2}h_{i-1/2,j+1}}{3\Delta x\Delta y}u_{i-1,j+1}^{*}+\frac{h_{i,j+1/2}h_{i+1/2,j}}{3\Delta x\Delta y}u_{i,j}^{*}-\frac{h_{i,j+1/2}h_{i-1/2,j}}{3\Delta x\Delta y}u_{i-1,j}^{*} \quad \text{(A11)}$$

Using the finite difference equations (A1) and (A5)–(A11) based on the flowchart shown in Figure A2, we numerically simulated the tsunami propagation.

The tsunami inundation was included as a moving boundary condition between land (dry cell) and sea (wet cell). We followed the method of *Iwasaki and Mano* [1979]. If the water height of the wet cell was higher than the height of the dry cell adjacent to the wet cell, the water flowing into the land cell changes the dry cell into a wet cell, where we calculate the vertically averaged horizontal velocity flow into the dry cell from the wet cell based on Eqs. (A5) – (A11). We assumed that the wet cell became a dry cell if the water thickness d is less than 1×10^{-4} m.

Acknowledgments

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